miniRT

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Contents

1 Linear algebra **2**
1.1 Vector . . . 2
1.2 Matrix . . . 2

2 Vector space objects **2**
2.1 Ray . . . 2
2.2 Plane . . . 2
2.3 Sphere . . . 2
2.4 Circle . . . 2
2.5 Cylinder . . . 3
2.6 Cone . . . 3

3 Raytracer **3**
3.1 Camera . . . 3
3.2 Ray intersections . . . 4
3.2.1 Plane . . . 4
3.2.2 Circle . . . 4
3.2.3 Sphere . . . 4
3.2.4 Cylinder . . . 4
3.2.5 Cone . . . 4
3.3 Hit point informations . . . 5
3.3.1 Surface normal . . . 5
3.3.2 UV coordinate . . . 5
3.4 Surface lightning . . . 5
3.5 Projected shadows . . . 5
3.6 Visual textures . . . 5
3.7 Bumping textures . . . 5

1 Linear algebra

1.1 Vector

A vector is a mathematical object described by it's components, it has 1 component for each dimension, so a 3D vector has 3 components.
It has the following operations for 2 vectors, $\vec{v}$, $\vec{w} \in \mathbb{R}^3$ and a scalar $\lambda \in \mathbb{R}$.

Addition	$\vec{v} + \vec{w}$	$=$	$(v_x + w_x, v_y + w_y, v_z + w_z)$
Substraction	$\vec{v} - \vec{w}$	$=$	$(v_x - w_x, v_y - w_y, v_z - w_z)$
Scale	$\lambda\vec{v}$	$=$	$(v_x \cdot \lambda, v_y \cdot \lambda, v_z \cdot \lambda)$
Dot product	$\vec{v} \cdot \vec{w}$	$=$	$v_x w_x + v_y w_y + v_z w_z$
Inner-product	$\langle \vec{v}, \vec{w} \rangle$	$=$	$\vec{v} \cdot \vec{w}$
Cross product	$\vec{v} \times \vec{w}$	$=$	$(v_y w_z - v_z w_y, v_z w_x - v_x w_z, v_x w_y - v_y w_x)$
Magnitude	$\|\|\vec{v}\|\|$	$=$	$\sqrt{v_x^2 + v_y^2 + v_z^2}$
Normalization	$\hat{v}$	$=$	$\vec{v} \,/\, \|\|\vec{v}\|\|$

1.2 Matrix

2 Vector space objects

2.1 Ray

For a point $o \in \mathbb{R}^3$, the origin of the ray, a vector $\hat{d} \in \mathbb{R}^3$, the normalalized direction vector of the ray, and a scalar for the direction vector, $t \in \mathbb{R}$. We define the normal equation of a line as :

$$R(t) = o + t\hat{d}$$

2.2 Plane

For a point $p \in \mathbb{R}^3$, the origin of the plane, a vector $\hat{v} \in \mathbb{R}^3$, the normalized orthogonal vector of the plane and a arbitrary point $x \in \mathbb{R}^3$, it is a point of the plane if :

$$(x - p) \cdot \hat{v} = 0$$

2.3 Sphere

For a point $p \in \mathbb{R}^3$ the origin of the sphere, a scalar $r \in \mathbb{R}$, the radius of the sphere and an arbitrary point $x \in \mathbb{R}^3$., it is a point of the sphere if :

$$||x - p||^2 = r^2$$

2.4 Circle

For a point $p \in \mathbb{R}^3$, the origin of the circle, a vector $\hat{v} \in \mathbb{R}^3$, the normalized orthogonal vector of the circle and a arbitrary point $x \in \mathbb{R}^3$, it is a point of the circle if $||x - p|| \leq r|$:

$$(x - p) \cdot \hat{v} = 0$$

$$||(x - p)|| \leq r$$

2.5 Cylinder

For a point $p \in \mathbb{R}^3$ the origin of the sphere, a a vector $\hat{v} \in \mathbb{R}^3$ the normalized direction vector, two scalars r and $h \in \mathbb{R}$, respectively the radius and the height of the cylinder. We define the normal equation of an infinite cylinder as :

$$||(x-p) - ((x-p) \cdot \hat{v})\hat{v}||^2 = r^2$$

Limit the height with :

$$-\frac{h}{2} \leq (r(t) - c) \cdot \hat{d} \leq \frac{h}{2}$$

And then use the previous formula for a circle to set the top and bottom cap of the cylinder.

2.6 Cone

For a point $p \in \mathbb{R}^3$ the origin of the cone, a a vector $\hat{v} \in \mathbb{R}^3$ the normalized direction vector. We define equation of an infinite cone as :

$$((x-p) \cdot \hat{v})^2 = \cos^2(\theta)||x-c||^2$$

3 Raytracer

3.1 Camera

Given informations :

- $\mathbf{C_{pos}}$: 3D Position as a point.
- $\mathbf{C_{dir}}$: 3D Direction as a normalized vector.
- fov : Vertical field of view in degrees, ranging from 0° to 180°.

Deduced informations :

- $\mathbf{C_{fov}}$: Vertical field of view in radians, ranging from 0 to π radians. We can define it with the following fomula :

$$\mathrm{C_{fov}} = \mathrm{fov} \cdot \frac{\pi}{180^\circ}$$

- $\mathbf{C_{flen}}$: Focal length that can be calculated as follows :

$$\mathrm{C_{flen}} = \tan\left(\frac{\mathrm{C_{fov}}}{2}\right)$$

- $\mathbf{S_{ratio}}$: Pixel screen ratio defined by $\dfrac{\text{Width}}{\text{Height}}$

Calculate a ray passing from the camera's origin and a pixel of the screen :

$$\mathbf{S_{dir}} = \left[\left(2\left(\frac{x+0.5}{\text{Width}}\right) - 1\right)\mathrm{S_{ratio}C_{flen}}, \left(1 - 2\left(\frac{y+0.5}{\text{Height}}\right) - 1\right)\mathrm{C_{flen}}, 1\right]$$

Calculate camera's right and up vector :

$$\mathbf{C_{right}} = (0,1,0) \times \mathrm{C_{dir}} \text{ and } \mathbf{C_{up}} = \mathrm{C_{dir}} \times \mathrm{C_{right}}$$

Get the ray shot through the given screen coordinate in space where $\mathbf{R_{pos}} = \mathrm{C_{pos}}$ and

$$\mathbf{R_{dir}} = \begin{bmatrix} \hat{\mathrm{C}}_{\mathrm{right}_x} & \hat{\mathrm{C}}_{\mathrm{up}_x} & \hat{\mathrm{C}}_{\mathrm{dir}_x} \\ \hat{\mathrm{C}}_{\mathrm{right}_y} & \hat{\mathrm{C}}_{\mathrm{up}_y} & \hat{\mathrm{C}}_{\mathrm{dir}_y} \\ \hat{\mathrm{C}}_{\mathrm{right}_z} & \hat{\mathrm{C}}_{\mathrm{up}_z} & \hat{\mathrm{C}}_{\mathrm{dir}_z} \end{bmatrix} \begin{bmatrix} \mathrm{S}_{\mathrm{dir}_x} \\ \mathrm{S}_{\mathrm{dir}_y} \\ 1 \end{bmatrix}$$

3.2 Ray intersections

For the previously defined variables and object equations we can define the following intersection equations.
You can find the distance t of the nearest positive root with :

$$t = \min \left\{ \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \frac{-b + \sqrt{b^2 - 4ac}}{2a} \right\}_{t>0}$$

3.2.1 Plane

$$t = \frac{(p - o) \cdot \hat{v}}{\hat{d} \cdot \hat{v}}$$

t is the distance between the plane and the ray origin followed by it's direction vector if it is a finite result, and if $t < 0$ the plane lays behind the ray origin.

3.2.2 Circle

Same as a plane but where you have to check wether or not the hitpoint - circle origin vector is less or equal to the radius of the circle.

3.2.3 Sphere

If you substitute the equation of a ray into the one of the sphere you obtain a second degree equation :

$$||(o + t\hat{d}) - p||^2 = r^2 \iff (\hat{d} \cdot \hat{d})t^2 + 2((o - p) \cdot \hat{d})t + ((o - p) \cdot (o - p) - r^2) = 0$$

$$\begin{aligned} a &= \hat{d} \cdot \hat{d} \\ b &= (o - p) \cdot \hat{d} \\ c &= (o - p) \cdot (o - p) - r^2 \end{aligned}$$

3.2.4 Cylinder

If you substitute the equation of a ray into the one of the cylinder you obtain a second degree equation :

$$||((o+t\hat{d})-p)-(((o+t\hat{d})-p)\cdot\hat{v})\hat{v}||^2 = r^2 \iff (\hat{d}\cdot\hat{d})t^2+2((o-p)\cdot\hat{d})t+((o-p)\cdot(o-p)-r^2) = 0$$

$$\begin{aligned} a &= \hat{d} \cdot \hat{d} \\ b &= (o - p) \cdot \hat{d} \\ c &= (o - p) \cdot (o - p) - r^2 \end{aligned}$$

3.2.5 Cone

If you substitute the equation of a ray into the one of the cone you obtain a second degree equation :

$$((o + t\hat{d} - p) \cdot \hat{v})^2 = \cos^2(\theta)||o + t\hat{d} - c||^2 \iff$$

$$\begin{aligned} a &= (\hat{d} \cdot \hat{v})^2 - \cos^2(\theta) \\ b &= 2((\hat{d} \cdot \hat{v})((o - p) \cdot \hat{v}) - \cos^2(\theta)(\hat{d} \cdot (o - p))) \\ c &= ((o - p) \cdot \hat{v})^2 - \cos^2(\theta)||(o - p)||^s \end{aligned}$$

3.3 Hit point informations

3.3.1 Surface normal

3.3.2 UV coordinate

3.4 Surface lightning

3.5 Projected shadows

3.6 Visual textures

3.7 Bumping textures